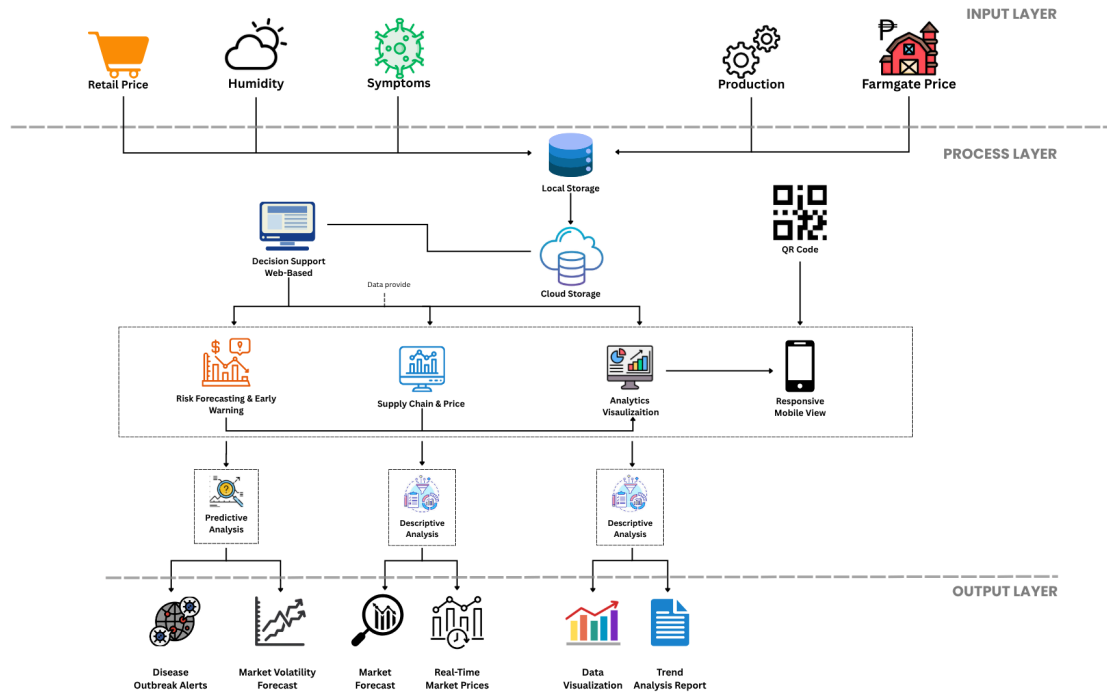


## Conceptual Framework

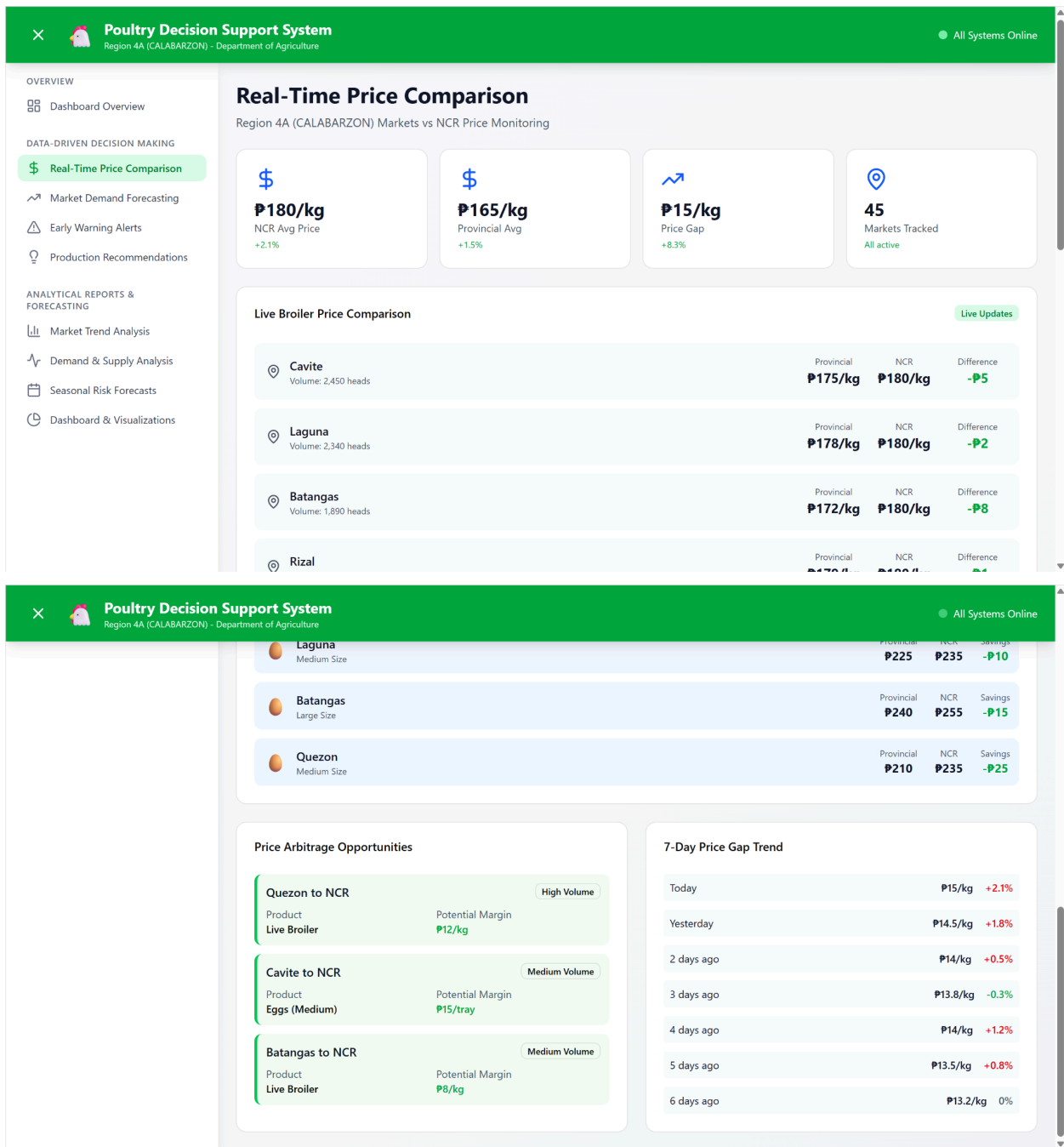


**Figure 1.1 Conceptual Framework of Proposed Study**

The conceptual framework demonstrates the steps involved in the Decision Support System for Poultry and Egg Stakeholders. The farmers insert data about their poultry farm condition such as humidity, symptoms, production, and farmgate prices through scanning on the auto generated QR code. The data is stored through cloud storage and local storage for backup. The web system that will be utilized by the Department of Agriculture consists of the main system which is the Decision Support System and the three proposed subsystems such as Risk Forecasting and Early Warning, Supply Chain and Price Monitoring, and Decision Support Dashboard. The data stored in the cloud storage will be obtained through the web system from cloud storage and processed through these subsystems. The Risk Forecasting and Early Warning is integrated with predictive analytics using Time-Series for its machine learning algorithm. The purpose is to predict the likelihood of having a disease in their poultry farm and forecasting the changes in the market, giving them an early warning on how to mitigate these challenges. The Supply Chain & Price Monitoring is used to monitor price monitoring between the NCR and Region IV A, providing the Department of Agriculture and farmers the price gap of the commodities between two regions. Lastly, the Decision Support Dashboard summarizes the information provided by two subsystems that will output recommendations for action and visualization using Decision Tree Algorithm. The output consists of disease outbreak, market

volatility, early warning and recommendation for action, market forecast, real-time market prices, and visualization which are the processed information that will be stored in the cloud and local storage then returned to the web system to be reflected in the dashboard of the farmers provided by the Department of Agriculture.

Subsystem Prototype



## Dashboard &amp; Visualizations

### Historical data analysis and demand predictions for Region 4A (CALABARZON)



**+12.5%**  
Projected Growth

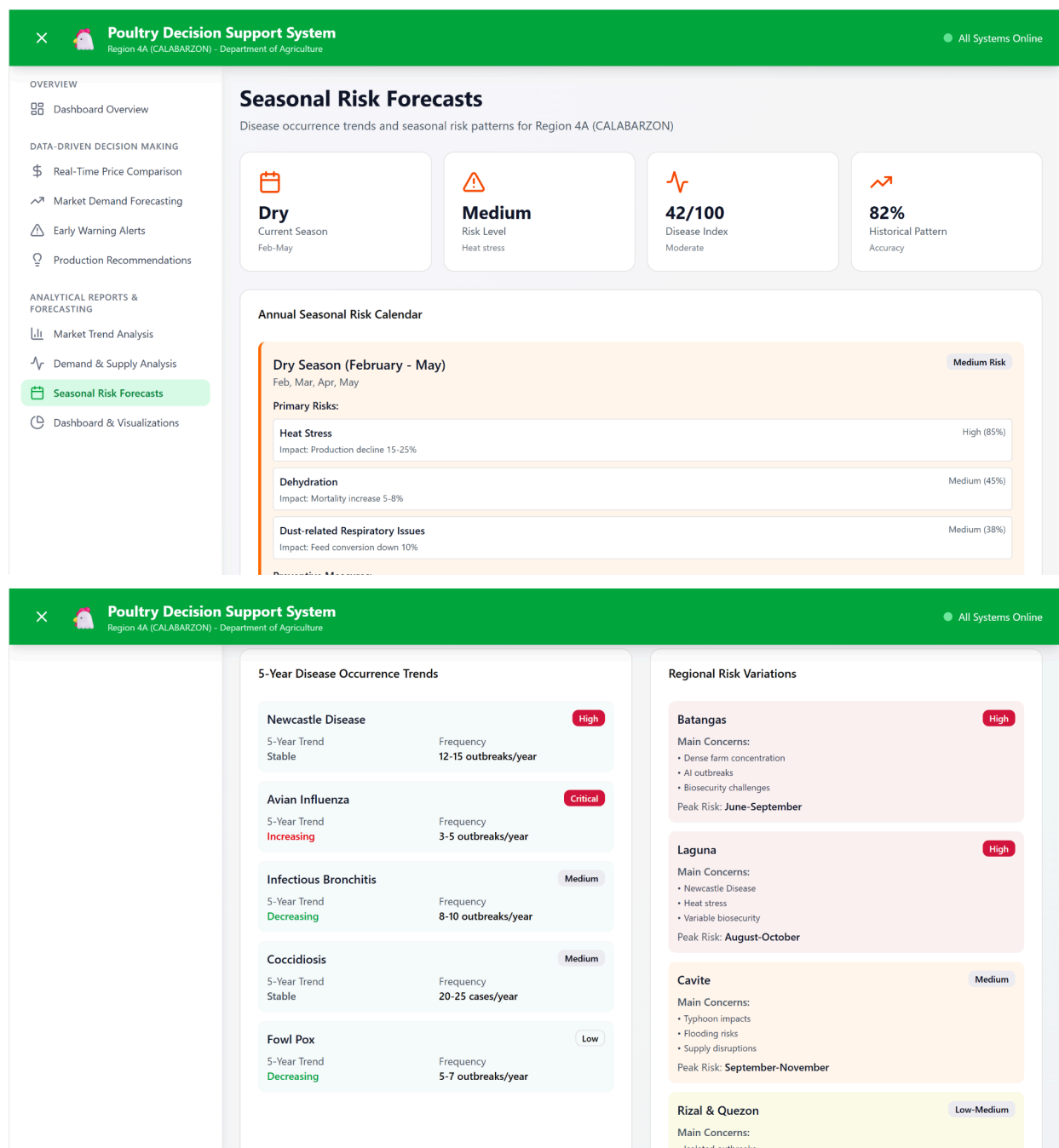
## +8%

### Production Recommendations Based on Forecast









The three subsystem prototypes embody the fundamental functionalities of the envisioned Poultry Decision Support System, each tailored to assist both poultry farmers and the Department of Agriculture in Region IV-A. These subsystems are integrated to furnish prompt market intelligence, forecast potential risks, and formulate well-informed recommendations, thereby facilitating improved decision-making processes.

The first subsystem, Supply Chain and Price Monitoring, is dedicated to monitoring current market dynamics and conducting price comparisons across various geographical areas.

This module allows stakeholders to view current price levels, market differences, price trends, and possible arbitrage opportunities. By consolidating supply chain and pricing data in a single interface, the subsystem helps farmers identify potentially more profitable sales opportunities and allows the DA to monitor market changes more effectively.

The second subsystem, Risk Forecasting and Early Warning, is designed to identify potential threats before they significantly harm poultry production. This includes monitoring disease outbreaks, seasonal risks, and other environmental factors that could disrupt farming operations. The module presents alert levels, regional risk status, and seasonal disease patterns, while also recommending preventive actions such as biosecurity measures and mitigation strategies. This feature provides stakeholders with advanced notice, potentially allowing them to mitigate losses and bolster their readiness.

The Decision Support Dashboard, the system's third subsystem, serves as its central analytical and advisory component. This element synthesizes data from market, supply chain, and risk forecasting modules to generate user-specific, actionable insights. Consequently, the dashboard offers production recommendations, trend analyses, and performance metrics, thereby assisting farmers in optimizing production levels, managing feed consumption, and improving operational planning. This subsystem effectively converts raw data into practical guidance, thereby facilitating both immediate and strategic decision-making processes.

## **Conclusion**

The conceptual framework demonstrates that farmers and the Department of Agriculture frequently exchange a great deal of information. Farmers input real-time data on their farms on their phones. The DA then compiles this information with information on local markets, sickness reports, and environmental data. The Decision Support Dashboard, Risk Forecasting and Early Warning, and Supply Chain and Price Monitoring are the three primary components of the system that collaborate to manage the data. The concepts and recommendations that can be implemented are the result of this process.

The framework makes it evident that the suggested system is intended to do more than simply store data. Additionally, it helps the poultry and egg industry make decisions by analyzing patterns and forecasting hazards. By utilizing Time-Series analysis and descriptive analytics, the approach can assist people in anticipating issues, comprehending how the market moves, and taking more effective action.